In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import yfinance as yf

from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout

df = yf.download(
    "AAPL",
    start="2015-01-01",
    end="2025-01-01",
    auto_adjust=False
)

print("Dataset loaded successfully!")
print("\nFirst 5 rows:")
print(df.head())

print("\nDataset shape:")
print(df.shape)

print("\nDataset Information:")
print(df.info())

print("\nMissing values:")
print(df.isnull().sum())

data = df[["Close"]].values

print("\nClose price data:")
print(data[:5])

plt.figure(figsize=(12, 6))

plt.plot(data)

plt.title("Apple (AAPL) Historical Closing Price")
plt.xlabel("Days")
plt.ylabel("Closing Price")

plt.show()

scaler = MinMaxScaler(feature_range=(0, 1))

scaled_data = scaler.fit_transform(data)

print("\nFirst 5 normalized values:")
print(scaled_data[:5])
```

```
time_step = 60

X = []
y = []

for i in range(time_step, len(scaled_data)):

    X.append(
        scaled_data[i - time_step:i, 0]
    )
    y.append(
        scaled_data[i, 0]
    )


X = np.array(X)
y = np.array(y)

print("\nSequence data:")
print("X shape:", X.shape)
print("y shape:", y.shape)

train_size = int(len(X) * 0.8)

X_train = X[:train_size]
X_test = X[train_size:]

y_train = y[:train_size]
y_test = y[train_size:]

print("\nTraining data:")
print("X_train:", X_train.shape)
print("y_train:", y_train.shape)

print("\nTesting data:")
print("X_test:", X_test.shape)
print("y_test:", y_test.shape)

X_train = X_train.reshape(
    X_train.shape[0],
    X_train.shape[1],
    1
)

X_test = X_test.reshape(
    X_test.shape[0],
    X_test.shape[1],
    1
)

print("\nAfter reshaping:")
```

```
print("X_train:", X_train.shape)
print("X_test:", X_test.shape)

model = Sequential()

model.add(
    LSTM(
        units=50,
        return_sequences=True,
        input_shape=(time_step, 1)
    )
)

model.add(
    Dropout(0.2)
)

model.add(
    LSTM(
        units=50,
        return_sequences=False
    )
)

model.add(
    Dropout(0.2)
)

model.add(
    Dense(25)
)

model.add(
    Dense(1)
)


model.compile(
    optimizer="adam",
    loss="mean_squared_error"
)

print("\nModel Summary:")
model.summary()


history = model.fit(
    X_train,
    y_train,
    epochs=20,
    batch_size=32,
    validation_split=0.1,
```

```
    verbose=1
)

predicted_price = model.predict(X_test)

predicted_price = scaler.inverse_transform(
    predicted_price
)

actual_price = scaler.inverse_transform(
    y_test.reshape(-1, 1)
)

rmse = np.sqrt(
    mean_squared_error(
        actual_price,
        predicted_price
    )
)

mae = mean_absolute_error(
    actual_price,
    predicted_price
)


print("\n=====================================")
print("MODEL EVALUATION")
print("=====================================")

print("RMSE:", rmse)
print("MAE :", mae)

plt.figure(figsize=(12, 6))

plt.plot(
    actual_price,
    label="Actual Price"
)

plt.plot(
    predicted_price,
    label="Predicted Price"
)

plt.title(
    "AAPL Stock Price - Actual vs Predicted"
)

plt.xlabel("Time")
plt.ylabel("Stock Price")
```

```
plt.legend()

plt.show()

plt.figure(figsize=(10, 5))

plt.plot(
    history.history["loss"],
    label="Training Loss"
)

plt.plot(
    history.history["val_loss"],
    label="Validation Loss"
)

plt.title(
    "Training and Validation Loss"
)

plt.xlabel("Epoch")
plt.ylabel("Loss")

plt.legend()

plt.show()

last_60_days = scaled_data[-time_step:]

last_60_days = last_60_days.reshape(
    1,
    time_step,
    1
)
next_day_prediction = model.predict(
    last_60_days
)

next_day_prediction = scaler.inverse_transform(
    next_day_prediction
)


print("\n=======================================")
print("NEXT DAY STOCK PRICE FORECAST")
print("=======================================")

print(
    "Predicted next day's closing price:",
    float(next_day_prediction[0][0])
)
```

```
[**********************100%***********************]  1 of 1 completed
```

```
Dataset loaded successfully!

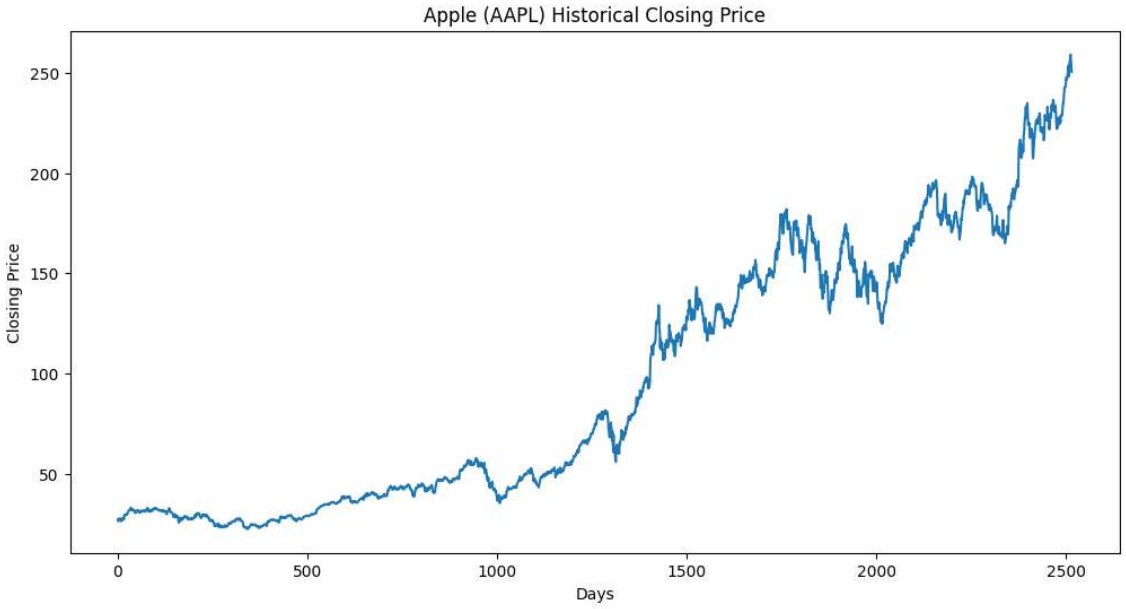
First 5 rows:
Price       Adj Close      Close       High        Low       Open     Volume
Ticker           AAPL       AAPL       AAPL       AAPL       AAPL       AAPL
Date
2015-01-02  24.171759  27.332500  27.860001  26.837500  27.847500  212818400
2015-01-05  23.490801  26.562500  27.162500  26.352501  27.072500  257142000
2015-01-06  23.493013  26.565001  26.857500  26.157499  26.635000  263188400
2015-01-07  23.822435  26.937500  27.049999  26.674999  26.799999  160423600
2015-01-08  24.737745  27.972500  28.037500  27.174999  27.307501  237458000

Dataset shape:
(2516, 6)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 2516 entries, 2015-01-02 to 2024-12-31
Data columns (total 6 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   (Adj Close, AAPL)  2516 non-null   float64
 1   (Close, AAPL)      2516 non-null   float64
 2   (High, AAPL)       2516 non-null   float64
 3   (Low, AAPL)        2516 non-null   float64
 4   (Open, AAPL)       2516 non-null   float64
 5   (Volume, AAPL)     2516 non-null   int64
dtypes: float64(5), int64(1)
memory usage: 137.6 KB
None

Missing values:
Price      Ticker
Adj Close  AAPL      0
Close      AAPL      0
High       AAPL      0
Low        AAPL      0
Open       AAPL      0
Volume     AAPL      0
dtype: int64

Close price data:
[[27.33250046]
 [26.5625    ]
 [26.56500053]
 [26.9375    ]
 [27.97249985]]
```




```
First 5 normalized values:
[[0.02007952]
 [0.01682281]
 [0.01683339]
 [0.01840887]
 [0.02278639]]

Sequence data:
X shape: (2456, 60)
y shape: (2456,)

Training data:
X_train: (1964, 60)
y_train: (1964,)

Testing data:
X_test: (492, 60)
y_test: (492,)

After reshaping:
X_train: (1964, 60, 1)
X_test: (492, 60, 1)

Model Summary:
```

```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/rnn/rnn.py:199:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer.
When using Sequential models, prefer using an `Input(shape)` object as the
first layer in the model instead.
  super().__init__(**kwargs)
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #

lstm (LSTM)	(None, 60, 50)	10,400
dropout (Dropout)	(None, 60, 50)	0
lstm_1 (LSTM)	(None, 50)	20,200
dropout_1 (Dropout)	(None, 50)	0
dense (Dense)	(None, 25)	1,275
dense_1 (Dense)	(None, 1)	26

Total params: 31,901 (124.61 KB)

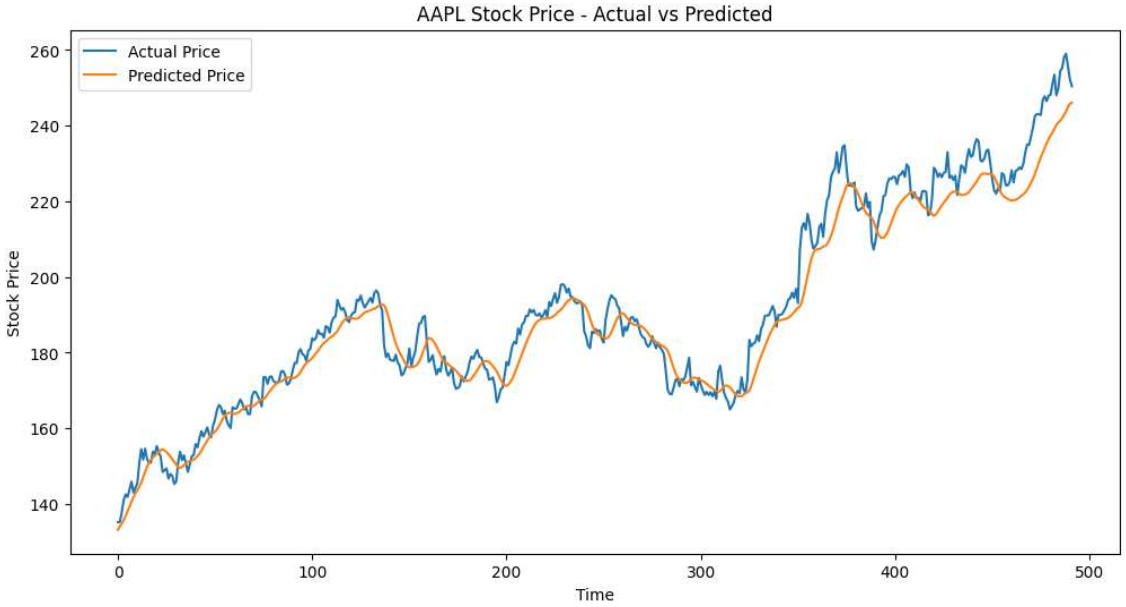
Trainable params: 31,901 (124.61 KB)

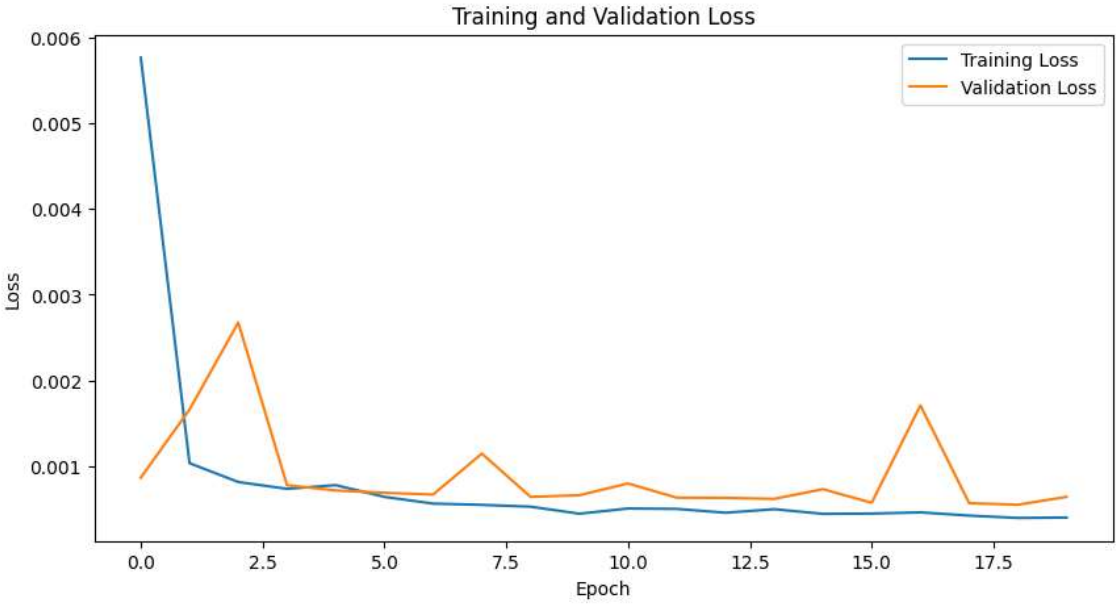
Non-trainable params: 0 (0.00 B)

```
Epoch 1/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 6s 53ms/step - loss: 0.0058 - val_loss: 8.6656e-04
Epoch 2/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 60ms/step - loss: 0.0010 - val_loss: 0.0017
Epoch 3/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 52ms/step - loss: 8.1606e-04 - val_loss: 0.0027
Epoch 4/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - loss: 7.3653e-04 - val_loss:
7.7919e-04
Epoch 5/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - loss: 7.7961e-04 - val_loss:
7.1767e-04
Epoch 6/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - loss: 6.4273e-04 - val_loss:
6.9034e-04
Epoch 7/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 4s 68ms/step - loss: 5.6429e-04 - val_loss:
6.7016e-04
Epoch 8/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - loss: 5.4983e-04 - val_loss: 0.0011
Epoch 9/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - loss: 5.2833e-04 - val_loss:
6.4255e-04
Epoch 10/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - loss: 4.4666e-04 - val_loss:
6.6198e-04
Epoch 11/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - loss: 5.0722e-04 - val_loss:
7.9911e-04
Epoch 12/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 5s 45ms/step - loss: 5.0251e-04 - val_loss:
6.3330e-04
Epoch 13/20
```

```
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - loss: 4.5797e-04 - val_loss:
6.3105e-04
Epoch 14/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - loss: 4.9925e-04 - val_loss:
6.1956e-04
Epoch 15/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 4s 68ms/step - loss: 4.4484e-04 - val_loss:
7.3245e-04
Epoch 16/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 53ms/step - loss: 4.4802e-04 - val_loss:
5.7404e-04
Epoch 17/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - loss: 4.6207e-04 - val_loss: 0.0017
Epoch 18/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - loss: 4.2454e-04 - val_loss:
5.6954e-04
Epoch 19/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - loss: 3.9730e-04 - val_loss:
5.5085e-04
Epoch 20/20
56/56 ━━━━━━━━━━━━━━━━━━━━ 4s 70ms/step - loss: 4.0185e-04 - val_loss:
6.4329e-04
16/16 ━━━━━━━━━━━━━━━━━━━━ 1s 31ms/step

========================================
MODEL EVALUATION
========================================
RMSE: 6.058068086031764
MAE : 4.7102141031404825
```






```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step

========================================
NEXT DAY STOCK PRICE FORECAST
========================================
Predicted next day's closing price: 246.02366638183594
```